Learning Topological Phases from Quantum Geometry

Machine Learning of Quantum Geometric Tensor Fields

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Progress Update-1

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1. Base paper and how it inspired us

Base paper: A.-K. Wu, L. Primeau, J. Zhang, K. Sun, Y. Zhang, S.-Z. Lin, "Modeling quantum geometry for fractional Chern insulators with unsupervised learning," *npj Comput. Mater.* **12**, 280 (2026), doi:10.1038/s41524-026-02155-1

What the paper does

- Learns form factors $\lambda_{\mathbf{q}}(\mathbf{k}) = \langle u_{\mathbf{k}} | u_{\mathbf{k}+\mathbf{q}} \rangle$ with a VAE and PCA.
- Data **generated from a physical Hamiltonian** (QBCP); no database.
- Latent space separates FCI from non-FCI *without labels*.
- PCA modes carry approximately quantized Chern numbers: **global topology is encoded in local geometry**.
- Gauge-invariant Ω, g via Wilson loops.

How it inspired our project

1. **Self-generated data** from analytic two-band Hamiltonians.
2. **BZ fields as images:** periodic-padded convolutions.
3. **Local → global:** our research question.
4. **Outlook (conv. + attention):** attention-pooling global branch.
5. **Gauge-free inputs:** built from $\mathbf{d}(\mathbf{k})$.

Where we differ: supervised and physics-informed (not unsupervised); single-particle Chern/$\mathbb{Z}_2$ topology (not many-body FCI); outputs are local QGT fields plus a topological class.

Research question: can a network infer global topology from local quantum-geometric fields?

Quantum geometric tensor (P = occupied-band projector)

$$Q_{ij} = \mathrm{Tr}[P\,\partial_i P\,\partial_j P] = g_{ij} - \tfrac{i}{2}\Omega_{ij}$$

Input $X_{\mathrm{loc}} \in \mathbb{R}^{19\times 40\times 40}$:
$\mathbf{d}, \hat{\mathbf{d}}, \partial_x\hat{\mathbf{d}}, \partial_y\hat{\mathbf{d}}, \nabla^2\hat{\mathbf{d}}, |\mathbf{d}|, |\mathbf{d}|^{-1}, |\partial_x\hat{\mathbf{d}}|, |\partial_y\hat{\mathbf{d}}|$
Output: $g_{xx}^{\mathrm{ML}}, g_{xy}^{\mathrm{ML}}, g_{yy}^{\mathrm{ML}}, \Omega_{xy}^{\mathrm{ML}}$ (shape $4\times 40\times 40$) and $C^{\mathrm{ML}} \in \{-1, 0, +1\}$.

Indexing Used:
ML = network prediction;
num = computed from $H(\mathbf{k})$;
ana = closed form.

What does our Data look like?

- **Synthetic, generated by our script** No QWZ dataset borrowed; no external database (JARVIS is the later stage).
- Family: randomized generalized two-band $H = \mathbf{d}(\mathbf{k})\cdot\boldsymbol{\sigma}$; contains QWZ as a special case.
- Exact analytic g_{ij}, Ω_{xy} and C labels.
- Tensors: X: $(225, 19, 40, 40)$, Y: $(225, 4, 40, 40)$.
- Balanced: 75 samples each for $C = -1, 0, +1$.
- Gap: min $= 0.403$, mean $= 1.009$ (cut ≥ 0.4).

`PhysicsQGTNet`

- Local branch + multi-scale periodic U-Net (3-level decoder).
- Attention-pooling global branch → Chern classifier.
- Circular padding on the BZ torus.

Physics-informed loss

$$\mathcal{L} = 4\mathcal{L}_{\mathrm{QGT}} + \mathcal{L}_{\mathrm{geo}} + 2\mathcal{L}_{\mathrm{topo}} + \tfrac{1}{2}\mathcal{L}_{\mathrm{cls}}$$

$\mathcal{L}_{\mathrm{geo}}$ enforces $\det g = \Omega_{xy}^2/4$;
$\mathcal{L}_{\mathrm{topo}}$ uses $C^{\mathrm{ML}} = \frac{1}{2\pi}\int \Omega_{xy}^{\mathrm{ML}} d^2k$.

Training log, 100 epochs (train metrics)

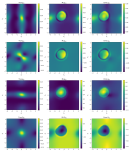
	Ep. 1	Ep. 40	Ep. 100
Total loss	6.036	0.00951	0.00193
$\mathcal{L}_{\mathrm{QGT}}$	0.0291	0.00161	0.00030
$\mathcal{L}_{\mathrm{geo}}$	0.0418	0.00311	0.00107
$\mathcal{L}_{\mathrm{topo}}$	0.4343	0.00019	0.00004
C-cls acc.	0.449	1.000	1.000
C-int MAE	0.555	0.0075	0.0030

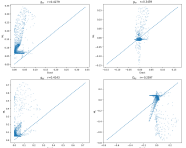
4. QWZ benchmark: analytical reference vs. ML

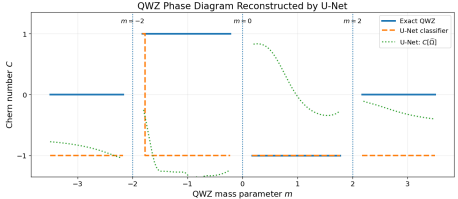
Exact QWZ scan, boundaries $m = -2, 0, +2$

Metric	Value
Classifier accuracy	0.466
Classifier macro-F1	0.274
Integrated-C MAE	0.733
Integrated-C RMSE	0.935
Mean field RMSE	0.299
Val. QGT MSE	0.0401
Val. round-C acc.	0.667

Accuracy 0.466 vs. 0.333 chance (3 classes): the epoch-1 model does not yet resolve the QWZ phase boundaries.







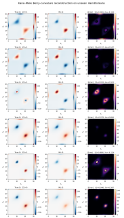
Reconstructed $\Omega_{xy}^{\mathrm{ML}}$, goodness-of-fit, QWZ phase diagram.

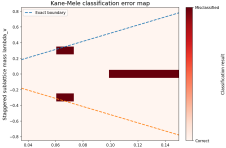
Unseen $C = 0$ Hamiltonian (min gap 2.452)

Field	MSE	MAE	MAE/std
g_{xx}	1.78×10^{-2}	0.111	6.63
g_{xy}	2.00×10^{-3}	0.0216	2.44
g_{yy}	1.70×10^{-2}	0.0869	2.41
$\mathrm{Im}\, Q_{xy}$	4.39×10^{-2}	0.114	4.39
Ω_{xy}	1.75×10^{-1}	0.229	4.39

Chern: exact 4.7×10^{-10}; $C_{\mathrm{int}}^{\mathrm{ML}} = -0.325$; classifier $= 0$ (correct).
Identity $\det g - \Omega^2/4$: true max 3.7×10^{-9}, predicted mean 0.035.

Kane–Mele ($\mathbb{Z}_2$)





Reconstruction and classification map. Result: $C^{\mathrm{ML}} \simeq -0.153$, $\mathbb{Z}_2^{\mathrm{ML}} = 0$.

The class label is correct, but MAE/std > 2 on every field and a 0.035 identity residual show the local geometry is not reconstructed faithfully.

6. Diagnosis and updated timeline

Diagnosis from output of the Model:

1. **Validation is not functioning:** $\mathrm{val} = 0.00000$ for all 100 epochs; final-validation metrics (acc. 0.449, C-int MAE 0.555, round-acc. 0.667) equal the epoch-1 train metrics to all printed digits.
2. Train metrics converge (C-cls $= 1.000$, $\mathcal{L}_{\mathrm{topo}} = 4 \times 10^{-5}$), but this is not evidence of generalization.

Completed (Oct 4)

- QWZ data/model and Kane Mele Pipeline.

Next (by Oct 27)

- Fix validation split/loss; select best checkpoint on held-out data.
- Retrain; rerun QWZ scan and unseen tests; regenerate all figures.
- Error vs. band gap; behaviour near phase boundaries.
- More Wannier Hamiltonians $\rightarrow$ multiband.

$\mathrm{QWZ} \rightarrow \mathrm{Kane\text{–}Mele} \rightarrow \mathrm{DFT/Wannier} \rightarrow \mathrm{multiband}$; **Presentation 3:** transferability + phase-boundary behaviour + interpretation.

Appendix A: Quantum geometric tensor

$$P(\mathbf{k}) = \sum_{n \in \mathrm{occ}} |u_n\rangle\langle u_n|, \qquad Q_{ij} = \mathrm{Tr}[P(\partial_i P)(\partial_j P)]$$

Two-band lower band:

$$g_{ij} = \tfrac{1}{4}\,\partial_i\hat{\mathbf{d}}\cdot\partial_j\hat{\mathbf{d}}, \quad \Omega_{xy} = -\tfrac{1}{2}\,\hat{\mathbf{d}}\cdot(\partial_x\hat{\mathbf{d}} \times \partial_y\hat{\mathbf{d}})$$

$$Q_{ij} = g_{ij} - \tfrac{i}{2}\Omega_{ij}, \quad g_{ij} = \mathrm{Re}\, Q_{ij}, \quad \Omega_{ij} = -2\,\mathrm{Im}\, Q_{ij}$$

Identity. $\det g = \frac{1}{16}|\partial_x\hat{\mathbf{d}} \times \partial_y\hat{\mathbf{d}}|^2$. Since $\partial_i\hat{\mathbf{d}} \perp \hat{\mathbf{d}}$, the cross product is parallel to $\hat{\mathbf{d}}$, so $|\partial_x\hat{\mathbf{d}} \times \partial_y\hat{\mathbf{d}}| = |2\Omega_{xy}|$:

$$\boxed{g_{xx}g_{yy} - g_{xy}^2 = \Omega_{xy}^2/4}$$

Sign of Ω_{xy} flips between bands. Exact for two-band pure states only.

Appendix B: QWZ model and phase boundaries

$$d_x = \sin k_x, \quad d_y = \sin k_y, \quad d_z = m + \cos k_x + \cos k_y$$

$$E_{\pm} = \pm\sqrt{\sin^2 k_x + \sin^2 k_y + (m + \cos k_x + \cos k_y)^2}$$

Band touching requires $d_x = d_y = d_z = 0$, hence $k_x, k_y \in \{0, \pi\}$.

$\mathbf{k}$	$(0,0)$	$(\pi,0)$	$(0,\pi)$	(π,π)
d_z	$m+2$	m	m	$m-2$

$$\boxed{m = -2,\ 0,\ +2}$$

$$\boxed{C^{\text{ana}} = \begin{cases} 0, & m < -2 \\ +1, & -2 < m < 0 \\ -1, & 0 < m < 2 \\ 0, & m > 2 \end{cases}}$$

ML phase boundaries should coincide with $m = -2, 0, +2$.

Appendix C: Kane–Mele phase boundaries

Valley $\eta = \pm 1$, spin $s = \pm 1$:

$$H_{\eta s} = v_F(\eta k_x \sigma_x + k_y \sigma_y) + m_{\eta s}\sigma_z$$

$$\boxed{m_{\eta s} = \lambda_v - \eta s\, 3\sqrt{3}\,\lambda_{\mathrm{SO}}}$$

$$E_{\eta s,\pm} = \pm\sqrt{v_F^2 k^2 + m_{\eta s}^2}$$

Transition: $m_{\eta s} = 0$.

$$\boxed{|\lambda_v| = 3\sqrt{3}\,|\lambda_{\mathrm{SO}}|}$$

$$\boxed{\mathbb{Z}_2^{\mathrm{ana}} = \begin{cases} 1, & |\lambda_v| < 3\sqrt{3}|\lambda_{\mathrm{SO}}| \\ 0, & |\lambda_v| > 3\sqrt{3}|\lambda_{\mathrm{SO}}| \end{cases}}$$

Gap closes and reopens at K/K'.

Appendix D: BHZ model and topological criterion

Basis $\{|E1,+\rangle, |H1,+\rangle, |E1,-\rangle, |H1,-\rangle\}$; TRS gives

$$H_{\mathrm{BHZ}} = \begin{pmatrix} h(\mathbf{k}) & 0 \\ 0 & h^*(-\mathbf{k}) \end{pmatrix}$$

$$h(\mathbf{k}) = \epsilon(k)\mathbb{1} + Ak_x\sigma_x - Ak_y\sigma_y + M(k)\sigma_z$$

$$\epsilon = C - Dk^2, \qquad M = M_0 - Bk^2$$

$$E_{\pm} = \epsilon \pm \sqrt{A^2k^2 + (M_0 - Bk^2)^2}$$

Gap closes at $k = 0$ when $\boxed{M_0 = 0}$.

Mass sign reversal $\operatorname{sgn} M(0) \neq \operatorname{sgn} M(\infty)$:

$$\frac{M_0}{B} > 0 \Rightarrow \text{topological}, \qquad \frac{M_0}{B} < 0 \Rightarrow \text{trivial}$$

$C_\uparrow = -C_\downarrow$, $C_{\mathrm{total}} = 0$, while $\mathbb{Z}_2$ can be non-trivial.

Appendix E: Phase boundaries at a glance

QWZ

$$\boxed{m = -2, 0, +2}$$

$$C^{\mathrm{ana}} = \begin{cases} 0 & m < -2 \\ +1 & -2 < m < 0 \\ -1 & 0 < m < 2 \\ 0 & m > 2 \end{cases}$$

Kane–Mele

$$\boxed{|\lambda_v| = 3\sqrt{3}|\lambda_{\mathrm{SO}}|}$$

$$\mathbb{Z}_2^{\mathrm{ana}} = \begin{cases} 1 & |\lambda_v| < 3\sqrt{3}|\lambda_{\mathrm{SO}}| \\ 0 & \text{otherwise} \end{cases}$$

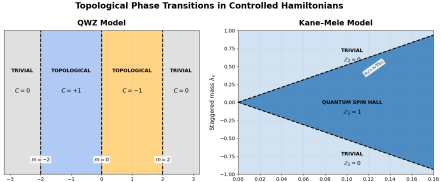
BHZ

$$\boxed{M_0 = 0}$$

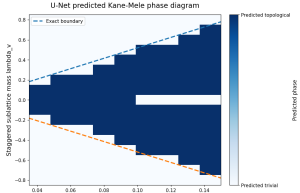
$\frac{M_0}{B} > 0 \Rightarrow$ topological

$\frac{M_0}{B} < 0 \Rightarrow$ trivial

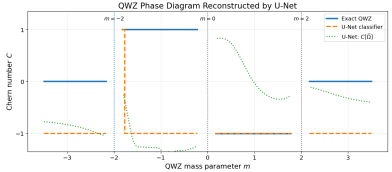
Appendix F: Phase-transition figures



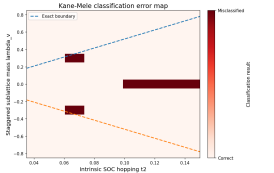
QWZ and Kane–Mele transition regimes.



Kane–Mele U-Net (ML) phase reconstruction.



QWZ reconstructed (ML) phase diagram.



Kane–Mele classification ($\mathbb{Z}_2^{\mathrm{ML}}$).

Appendix G: Implementation summary

Training family (14 random coefficients):

$$d_x = a_1 \sin k_x + a_2 \sin k_y + a_3 \sin(k_x + k_y) + a_4 \sin 2k_x, \quad d_y = b_1 \sin k_y + b_2 \sin k_x + b_3 \sin(k_x - k_y) + b_4 \sin 2k_y,$$

$$d_z = M + \sum_{j=1}^{6} c_j \cos(\cdot); \qquad \text{QWZ: } a_1 = b_1 = c_1 = c_2 = 1, \text{ others } 0$$

Loss terms

$$\mathcal{L}_{\text{geo}} = \left\langle \log\left(1 + \left| \det g^{\text{ML}} - \tfrac{1}{4}(\Omega_{xy}^{\text{ML}})^2 \right|\right) \right\rangle$$

$$\mathcal{L}_{\text{topo}} = \text{SmoothL1}(C^{\text{ML}}, C), \quad C^{\text{ML}} = \tfrac{1}{2\pi} \sum_{\mathbf{k}} \Omega_{xy}^{\text{ML}} \delta k^2$$

$\mathcal{L}_{\text{cls}}$: cross-entropy over $C \in \{-1, 0, +1\}$.

Network

- Periodic padding; multi-scale dilated blocks; 3-level U-Net decoder.
- High-res. local branch fused before QGT head.
- Attention pooling $\rightarrow$ Chern head.

Benchmark: exact QWZ, 141 values of $m \in [-3.5, 3.5]$.

Training milestones (train metrics)

Epoch	Total	$\mathcal{L}_{\mathrm{QGT}}$	C-int MAE
1	6.036	0.0291	0.555
10	0.0715	0.0102	0.0387
20	0.0723	0.0081	0.0650
50	0.0655	0.0026	0.0853
100	0.00193	0.00030	0.0030

Validation loss = 0.00000 in all epochs.

Full QGT at $\mathbf{k} = (0, 0)$, unseen Hamiltonian

$$Q^{\mathrm{true}} = \begin{pmatrix} 0.0997 & 0.0291 - 0.1603i \\ 0.0291 + 0.1603i & 0.2660 \end{pmatrix}$$

$$Q^{\mathrm{ML}} = \begin{pmatrix} 0.0831 & -0.0044 - 0.0600i \\ -0.0044 + 0.0600i & 0.0550 \end{pmatrix}$$

Off-diagonal imaginary part (curvature) is underestimated by a factor ≈ 2.7.

Appendix I: References

- **[Base paper]** A.-K. Wu, L. Primeau, J. Zhang, K. Sun, Y. Zhang and S.-Z. Lin, "Modeling quantum geometry for fractional Chern insulators with unsupervised learning," *npj Comput. Mater.* **12**, 280 (2026), doi:10.1038/s41524-026-02155-1
- X.-L. Qi, Y.-S. Wu and S.-C. Zhang, *Phys. Rev. B* **74**, 085308 (2006).
- C. L. Kane and E. J. Mele, *Phys. Rev. Lett.* **95**, 146802 (2005).
- C. L. Kane and E. J. Mele, *Phys. Rev. Lett.* **95**, 226801 (2005).
- B. A. Bernevig, T. L. Hughes and S.-C. Zhang, *Science* **314**, 1757 (2006).
- NIST JARVIS-DFT database.